1. Koblitz $m = 2…8$ high-arity correctness inputs

2026-10-18. Planted chains and point-specialized direct polynomials; no natural solver or ECDLP cost result.

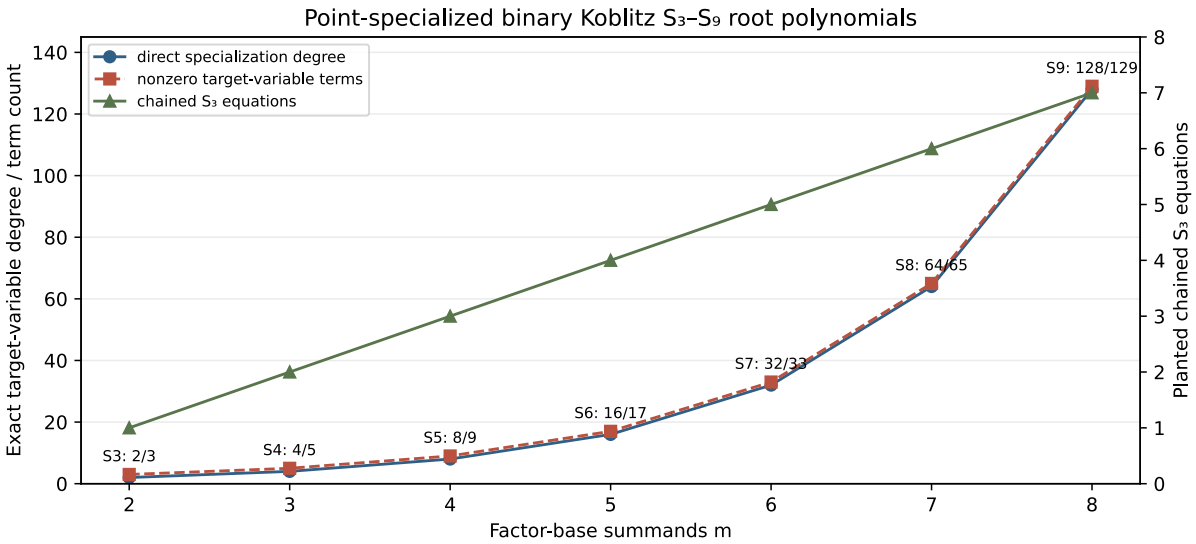
1.1. Fixed inputs

The certified curve is $y^2 + xy = x^3 + x^2 + 1$ over `GF(2^73)` with a 62.224-bit prime subgroup. The eight predetermined base-point indices are `0,73,146,219,292,365,1,74` in the independently replayed 438-coordinate six-orbit base. Each m-term target is the full-point sum of the first m selected points.

The characteristic-two equation is $S_3(x, y, z) = (xy + xz + yz)^2 + xyz + 1$. Every planted chain has $m - 1$ checked equations. For the direct control, fix all m base coordinates, enumerate the 2^m sign assignments, fold global point negation, and form the monic polynomial in target x from the resulting 2^{m-1} roots. An independent Sage replay checks the full points, target sums, roots, coefficients, nonroot controls and the factorized recursive resultant at every arity.

1.2. Exact specialization sizes

Summands	Direct label	Degree / nonzero terms	Chain S_3 equations
2	S_3	2 / 3	1
3	S_4	4 / 5	2
4	S_5	8 / 9	3
5	S_6	16 / 17	4
6	S_7	32 / 33	5
7	S_8	64 / 65	6
8	S_9	128 / 129	7



All coefficients of these univariate specializations are nonzero on the fixed points. Direct S_8 addresses seven summands plus target; eight summands require S_9 or a chain. The producer's construction wall was 0.379315 seconds after separate factor-base preparation. The first Sage replay attempt failed only at JSON serialization of a Sage integer and its partial bytes are retained; the corrected complete replay passed.

1.3. Interpretation and next gate

Fixing every base coordinate makes these target-variable polynomials tractable. It does not construct full multivariate S_8 or S_9, measure solver degree, certify natural relations, or compare complete costs. The next frozen comparison requires same-size base controls, symbolic direct/chained formulations, planted and negative solver checks, then natural yield, distinct rank, linear algebra, unseen-target work, charged setup and matched rho. Unexecuted cells remain unavailable, and resource stops remain censored.

Provenance: Full sources, exact bytes and failure scope are in `docs/reports/2026-10-18-koblitz-high-arity-inputs.md` and `research/icperf_koblitz_high_arity_20261018/`. This input screen changes no canonical attack-cost graph edge.